Sea-Bird Scientific
13431 NE 20th Street
Bellevue, WA 98005
USA

+1 425-643-9866
seabird@seabird.com
www.seabird.com

SENSOR SERIAL NUMBER: 9431
CALIBRATION DATE: 10-Mar-21

Slocum Payload CTD CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.001938e+000
h =  1.317039e-001
i = -1.733771e-004
j =  3.007991e-005
```

```
CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  1.5091e-007
```

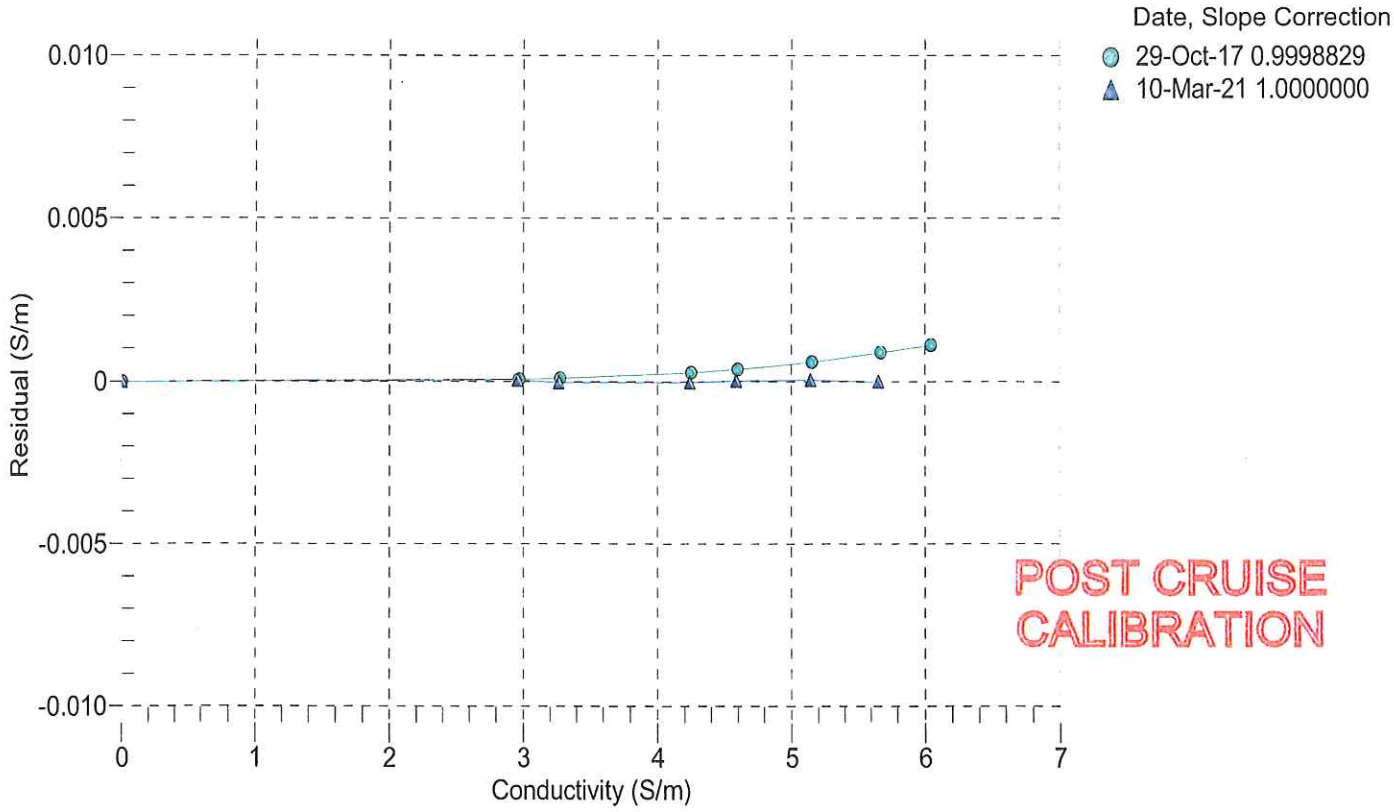
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2760.78	0.00000	0.00000
1.0000	34.5994	2.95919	5485.15	2.95922	0.00003
4.5000	34.5798	3.26460	5691.93	3.26457	-0.00003
15.0000	34.5385	4.24112	6307.11	4.24110	-0.00003
18.5000	34.5299	4.58446	6509.34	4.58447	0.00001
23.9999	34.5209	5.13951	6823.27	5.13954	0.00003
29.0000	34.5166	5.65873	7104.01	5.65871	-0.00002
32.4999	34.5136	6.02914	7297.48	6.02905	-0.00009

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C); p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



POST CRUISE CALIBRATION

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SENSOR SERIAL NUMBER: 9431
CALIBRATION DATE: 04-Mar-21

Slocum Payload CTD PRESSURE CALIBRATION DATA
1450 psia S/N 10712103

COEFFICIENTS:

PA0 =	7.084729e-001	PTCA0 =	5.246266e+005
PA1 =	4.446180e-003	PTCA1 =	1.185899e+001
PA2 =	-1.616291e-011	PTCA2 =	-2.149627e-001
PTEMPA0 =	-6.684982e+001	PTCB0 =	2.528650e+001
PTEMPA1 =	5.087789e-002	PTCB1 =	1.000000e-004
PTEMPA2 =	-4.094839e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.50	527895.5	1808.2	14.53	0.00
303.06	592452.1	1815.5	301.45	-0.11
588.73	657113.6	1816.9	588.71	-0.00
875.94	721787.0	1818.5	875.89	-0.00
1163.05	786460.9	1819.5	1162.93	-0.01
1450.22	851215.2	1821.0	1450.20	-0.00
1163.06	786515.1	1822.0	1163.17	0.01
875.84	721803.6	1822.6	875.96	0.01
588.77	657125.1	1823.6	588.76	-0.00
301.55	592458.2	1824.8	301.48	-0.01
14.49	527893.0	1828.4	14.51	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	1984	527927.80
29.00	1913	527939.80
24.00	1812	527939.60
18.50	1701	527920.60
15.00	1630	527899.00
4.50	1419	527818.20
1.00	1348	527789.80

TEMPERATURE (°C)	SPAN
-5.00	25.29
35.00	25.29

y = thermistor output (counts)

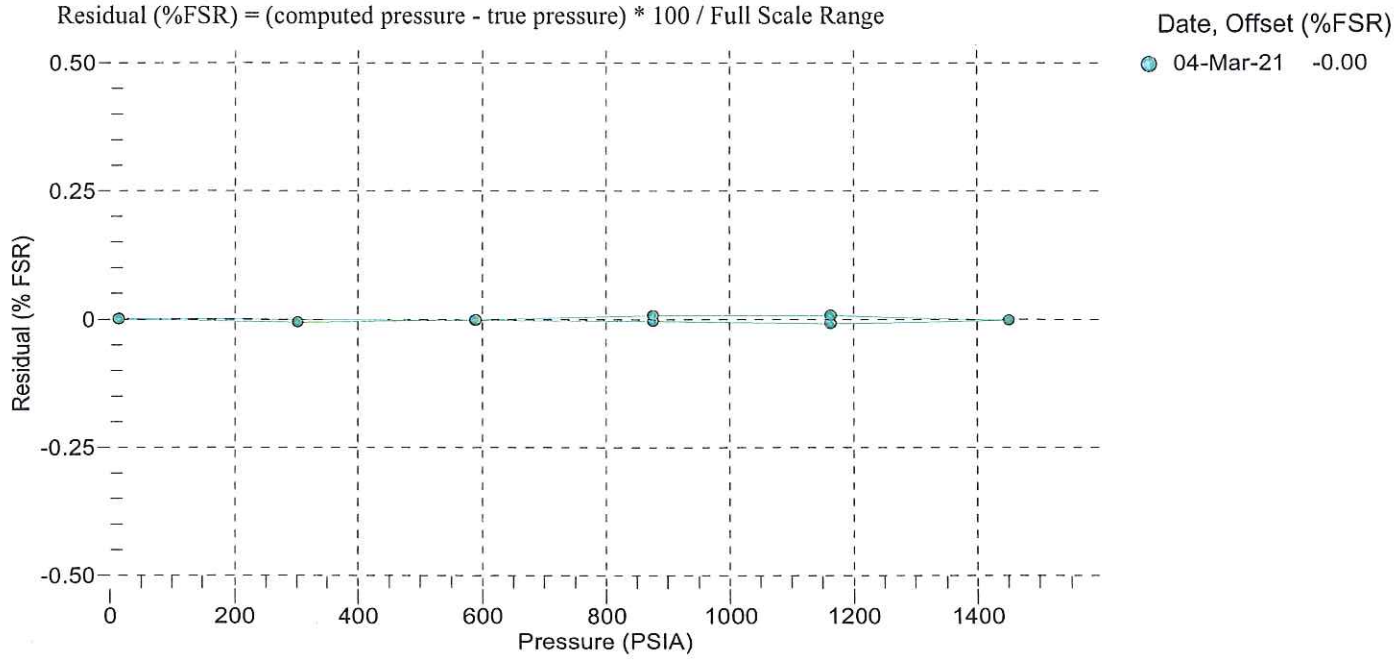
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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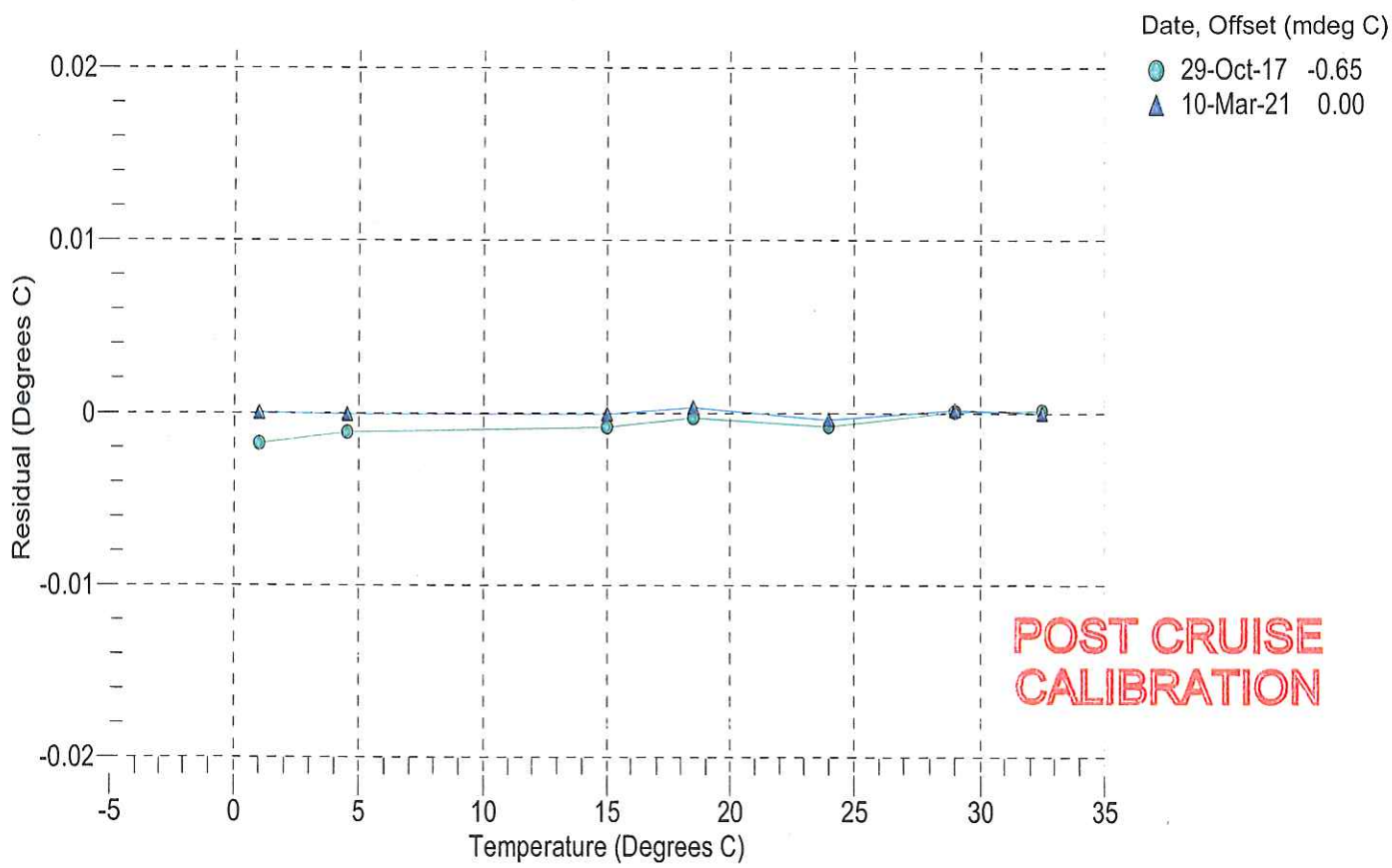
Slocum Payload CTD TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

COEFFICIENTS:

```
a0 = -2.183282e-004
a1 =  3.271953e-004
a2 = -5.874713e-006
a3 =  2.396599e-007
```

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	576738.8	1.0000	0.0000
4.5000	494094.8	4.4999	-0.0001
15.0000	316760.8	14.9999	-0.0001
18.5000	274820.8	18.5003	0.0003
23.9999	221167.2	23.9995	-0.0004
29.0000	182632.8	29.0002	0.0002
32.4999	160262.0	32.4999	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature